

Exercice : From transfer function to state-feedback control (2)

Let us consider a system described by the following transfer function :

$$\frac{Y}{U}(s) = \frac{s+1}{s(s+10)^2} \quad (1)$$

Part 1 : Pole Placement and State-Observer

The specifications are :

- overshoot : 5%
- time response : 2 seconds
- No steady-state error

We suppose that the state is not accessible, but only the output is measured.

- 1.1** Determine a state-feedback control of the following form that ensures the specifications (*done in previous WC2*)

$$u = -Kx + ly^r$$

- 1.2** According to the state-space representation you chose, is the system observable?

- 1.3** Propose to design a state-observer

- 1.4** Using the joint Matlab-Simulink files, implement this solution.

Part 2 : LQ control and State-observer

We now want to implement a control that minimizes

$$\int_t^{+\infty} (y(\tau)^2 + \alpha u^2(\tau)) d\tau$$

- 2.1** Suppose that $\alpha = 10^{-4}$. Determine the state-feedback control that ensures these new specifications

- 2.2** Propose an adequate state-observer

- 2.3** Using the joint Matlab-Simulink files, implement this new solution.

Part 3 : Noise and State-observer

We now assume that the output sensor is noisy, with centred white noise of variance dy_0 . Similarly, the actuator is noisy, with noise of variance du_0 .

- 3.1** Rewrite the state-space equations to take these noises into consideration

- 3.2** Check if the Kalman conditions are ensured to design a state-observer using the Kalman approach.

- 3.3** For different values of dy_0 and du_0 , determine the gain L of the state observer.

- 3.4** Using the joint Matlab-Simulink files, implement this solution, and compare it with the previous state-observer.